

4.89) THE GRADIENT IS A VECTOR. -7 IS A SCALAR. WE ARE MISSING THE DIRECTION INFORMATION.

4.95)

$$c) \vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}$$

$$\frac{\partial f}{\partial x} = 2x$$

$$\frac{\partial f}{\partial y} = 2y$$

$$\Rightarrow \vec{\nabla} f = 2x \hat{i} + 2y \hat{j}$$

$$\vec{\nabla} f(0,0) = 0$$

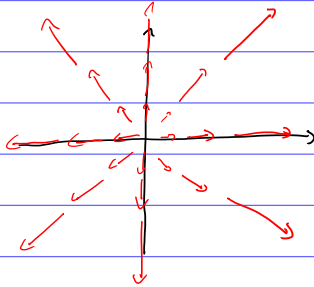
$$\vec{\nabla} f(0,1) = 2\hat{j}$$

$$\vec{\nabla} f(0,-1) = -2\hat{j}$$

$$\vec{\nabla} f(1,0) = 2\hat{i}$$

$$\vec{\nabla} f(-1,0) = -2\hat{i}$$

b)



POINTING OUTWARD AND INCREASING
IN MAGNITUDE AWAY FROM ZERO.

4.107)

$$a) \vec{\nabla} V = \frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j}$$

$$\begin{aligned} \frac{\partial V}{\partial x} &= \frac{-k}{2((x-x_0)^2 + y^2)^{3/2}} 2(x-x_0) + \frac{k}{2((x+x_0)^2 + y^2)^{3/2}} 2(x+x_0) \\ &= \frac{k(x+x_0)}{[(x+x_0)^2 + y^2]^{3/2}} - \frac{k(x-x_0)}{[(x-x_0)^2 + y^2]^{3/2}} \end{aligned}$$

$$\begin{aligned} \frac{\partial V}{\partial y} &= \frac{-k}{2[(x-x_0)^2 + y^2]^{3/2}} (2y) + \frac{k}{2[(x+x_0)^2 + y^2]^{3/2}} (2y) \\ &= \frac{y k}{[(x+x_0)^2 + y^2]^{3/2}} - \frac{y k}{[(x-x_0)^2 + y^2]^{3/2}} \end{aligned}$$

$$\begin{aligned} \Rightarrow \Sigma &= \left(\frac{k(x-x_0)}{[(x-x_0)^2 + y^2]^{3/2}} - \frac{k(x+x_0)}{[(x+x_0)^2 + y^2]^{3/2}} \right) \hat{i} \\ &+ \left(\frac{y k}{[(x-x_0)^2 + y^2]^{3/2}} - \frac{y k}{[(x+x_0)^2 + y^2]^{3/2}} \right) \hat{j} \end{aligned}$$

b) $x > x_0$ AND $y = 0$

$$\begin{aligned} \Rightarrow \Sigma &= \left(\frac{k(x-x_0)}{((x-x_0)^2)^{3/2}} - \frac{k(x+x_0)}{((x+x_0)^2)^{3/2}} \right) \hat{i} \\ &= k \left(\frac{1}{(x-x_0)^2} - \frac{1}{(x+x_0)^2} \right) \hat{i} \end{aligned}$$

IF $x > x_0$

$$\Rightarrow \frac{1}{(x-x_0)^2} > \frac{1}{(x+x_0)^2}$$

$$\Rightarrow \Sigma > 0$$

$\Rightarrow$ THE FORCE WILL BE IN THE
+VE DIRECTION

$$c) \Sigma (5x_0, 2x_0, 0) = h \left(\frac{4x_0}{((4x_0)^2 + (2x_0)^2)^{3/2}} - \frac{6x_0}{((6x_0)^2 + (2x_0)^2)^{3/2}} \right) \hat{i} \\ + h \left(\frac{2x_0}{((4x_0)^2 + (2x_0)^2)^{3/2}} - \frac{2x_0}{((6x_0)^2 + (2x_0)^2)^{3/2}} \right) \hat{j}$$

DENOMINATORS:

$$\begin{aligned} ((4x_0)^2 + (2x_0)^2)^{3/2} &= (16x_0^2 + 4x_0^2)^{3/2} \\ &= (20x_0^2)^{3/2} \\ &= 20^{3/2} x_0^3 \end{aligned}$$

$$\begin{aligned} ((6x_0)^2 + (2x_0)^2)^{3/2} &= (36x_0^2 + 4x_0^2)^{3/2} \\ &= (40x_0^2)^{3/2} \\ &= 40^{3/2} x_0^3 \end{aligned}$$

$$\Rightarrow \Sigma() = h \left(\frac{4}{20^{3/2} x_0^3} - \frac{6}{40^{3/2} x_0^3} \right) \hat{i} \\ + h \left(\frac{2}{20^{3/2} x_0^3} - \frac{2}{40^{3/2} x_0^3} \right) \hat{j}$$

$$= h A / x_0^3 \hat{i} + h B / x_0^3 \hat{j}$$

$$A = \frac{4}{20^{3/2}} - \frac{6}{40^{3/2}}$$

$$B = \frac{2}{20^{3/2}} - \frac{2}{40^{3/2}}$$

$$\hat{c} = \arctan \left(\frac{B}{A} \right) =$$

4.129)

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j}$$

$$\frac{\partial f}{\partial x} = 12x^2 + 6x - 90$$

$$\frac{\partial f}{\partial y} = 3y^2 - 48$$

$$\vec{\nabla} f = 0 \vec{i} + 0 \vec{j} \quad \text{FOR CRITICAL POINTS.}$$

$$\text{SOLVE FOR } 12x^2 + 6x - 90 = 0$$

$$\Rightarrow x = \frac{-6 \pm \sqrt{36 + 4 \times 90 \times 12}}{24} = \frac{-6 \pm 66}{24}$$

$$\Rightarrow x = -3 \text{ OR } 2.5$$

$$y = \pm 4$$

$$\boxed{\text{CRITICAL POINTS: } (-3, -4), (-3, 4), (2.5, -4), \text{ AND } (2.5, 4)}$$

FOR THE SECOND DERIVATIVE, USE

$$D(x, y) = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} (3y^2 - 48) = 0$$

$$\frac{\partial^2 f}{\partial x^2} = 24x + 6$$

$$\frac{\partial^2 f}{\partial y^2} = 12y$$

$$\text{AT } (-3, -4):$$

$$\frac{\partial^2 f}{\partial x^2} = -66$$

$$\frac{\partial^2 f}{\partial y^2} = -48$$

$\Rightarrow \text{MAX.}$

$$\text{AT } (-3, 4)$$

$$\frac{\partial^2 f}{\partial x^2} = -66$$

$$\frac{\partial^2 f}{\partial y^2} = 48$$

$\Rightarrow \text{NEITHER A MIN OR MAX.}$

$$\text{AT } (2.5, -4)$$

$$\frac{\partial^2 f}{\partial x^2} = 66$$

$$\frac{\partial^2 f}{\partial y^2} = -48$$

$\Rightarrow \text{NEITHER A MIN OR MAX}$

$$\text{AT } (2.5, 4)$$

$$\frac{\partial^2 f}{\partial x^2} = 66$$

$$\frac{\partial^2 f}{\partial y^2} = 48$$

$\Rightarrow \text{MIN.}$